

The “Difficulties” in Fermat’s Original Discourse on the Indecomposability of Powers Greater Than a Square: A Retrospect

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DOI: 10.13140/RG.2.2.24342.32321

June 1, 2026

Abstract

We present a reconstruction of Pierre Fermat’s possible manuscript reasoning based on the two-power decomposition constraint and on the mutual symmetry of two natural binomial scales.

Starting from a hypothetical positive natural solution of

$$x^n + y^n = z^n, \quad n > 2,$$

we use the symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

where m and p are first treated as positive real parameters. This gives

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of this binomial difference admits two natural real normalizations. The first extracts the linear binomial factor n :

$$y^n = 2n l^n.$$

The second extracts the total odd binomial coefficient mass 2^{n-1} :

$$y^n = 2^n q^n.$$

Here l and q are positive real residual scales. The symbol q denotes a scale parameter and must not be confused with a modulus in modular arithmetic.

Comparing the two representations gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

The central structural condition of the present reconstruction is the mutual scale-symmetry condition

$$l^n = q^n.$$

Under this condition one obtains

$$1 = \frac{2^{n-1}}{n},$$

or equivalently

$$n = 2^{n-1}.$$

This equation has exactly two positive integer solutions,

$$n \in \{1, 2\}.$$

Thus, under the mutual scale-symmetry condition, every hypothetical positive integer solution with $n > 2$ leads to a contradiction.

The decomposition premise is kept as an explicit structural assumption over $\mathbb{N}$. The real parametrization is used as an algebraic reconstruction, while the integer specialization, parity conditions, and modular caveat are treated separately. In particular, modular congruences are not identified with integer equalities. The accompanying Coq formalization verifies the final exponential-linear obstruction

$$n = 2^{n-1} \implies n \in \{1, 2\},$$

and related bridge lemmas, but the decisive mathematical point remains the status of the mutual scale-symmetry condition $l^n = q^n$.

Keywords: number theory, Fermat’s Big/Last Theorem, History of Mathematics, Algebra, Conditional proof

Classification code of ACM CSS 1998: D.2.4; F.3.1; F.4.1

Classification code of MSC: 68V20; Secondary: 68V15, 03B35

Competing interests

No, I declare that the authors have no competing interests as defined by Springer, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

The author(s) declare(s) that there are no conflicts of interest regarding the publication of this article.

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and <https://www.fermatslibrary.com/p/0c39c9be>

and posted as a preprint some small previous version (October 2023):

https://www.researchgate.net/publication/374350359_The_Difficulties_in_Fermat's_Original_Discourse_on_the_Indecomposability_of_Powers_Greater_Than_a_Square_A_Respect

and this current finished version as a preprint is published there at the same link.

The data for this finished article are also available at the following link (you can get all previous versions from this link): <https://doi.org/10.31219/osf.io/jbdas>

Funding

No funding

Ethics approval and consent to participate.

Not applicable

Authors’ contributions

The entire article was written by the first author alone. The acknowledgements are given in the Acknowledgements section.

Acknowledgements

First and foremost, the author expresses gratitude to Almighty God and His moral commandments, which have always supported the author in difficult moments.

The author also wishes to honor the memory of his mother, Natalia Alekseevna Dedenko, whose unwavering faith in him and his work has remained a source of inspiration even years after her passing. Her support and belief in the value of perseverance and intellectual pursuit continue to resonate in this research.

Additionally, the author expresses his deepest gratitude to his father, Leonid Grigoryevich Dedenko, Doctor of Physical and Mathematical Sciences, Professor at the Faculty of Physics of Moscow State University, for his invaluable guidance, mentorship, and foundational influence on the author's scientific development. The author is also grateful to mathematics teacher Kolesnikova V.E. from school No. 533 in Moscow for her early support in mathematical education.

The author expresses appreciation to associate professors of the National Research Nuclear University MEPhI Malov A.F., Ivliev S.V., Makarova I.F., Deputy Head of the Department of Applied Nuclear Physics of the NRNU MEPhI Ryabeva E.V., and IAEA staff member Yurkin P.G. for their insightful discussions and professional support.

Further appreciation is extended to Associate Professor Chernyshev L.N. from the Financial University under the Government of the Russian Federation, Dr. Boris Borisov from the Faculty of Mechanics and Mathematics of Moscow State University, Dr. Pashchenko F.F. and colleagues from IPU RAS, Associate Professor Miloserdin V.Yu. from NRNU MEPhI, and NRNU MEPhI teacher Mukhin V.I. for their valuable feedback and technical discussions. The author acknowledges Sergey P. Klykov, fermentation expert from Alpha-Integrum, Ltd, and Ph.D. Math Coordinator at Lebanese International University Issam Kaddoura for their contributions to certain analytical aspects of the work.

Acknowledgment is also given to the creators of generative neural networks—ChatGPT, Claude, Gemini, DeepSeek, Mistral—for their role in partially verifying and improving the results presented in this article. The author appreciates members of mathematical forums such as dxdy and cyberforum for their critical feedback, which enhanced the quality of the research. Sincere thanks are also due to the supervisor, Associate Professor Kadilin V.V. of NRNU MEPhI, Dean of the Physical-Technical Faculty of NRNU MEPhI Tikhomirov G.V., and Professors Samosadny V.T., Filippov V.P., Kramer-Ageev E.A., Moore V.D., Petrunin V.F., Bolozdyne A.I., Dmitrenko V.V., Grachev V.M., Mashkovich V.P., Ulin S.E., and Associate Professors Boyko N.V., Evstyuhina I.A., Kaplun A.A., Kutsenko K.V., Kolesnikov S.V., Minayev V.M., Petrov V.I., Samonov A.M., Sulaberidze G.A., and Rostokin V.I. from NRNU MEPhI for their academic support and valuable insights.

Further appreciation is given to Project Director in India of ATOM-STROYEXPORT JSC Angelov V.A., First Deputy Director for the Construction of KUDANKULAM NPP of ATOMSTROYEXPORT JSC Kvasha A.V., and colleagues from the KUDANKULAM NPP Project Management Department of ASE: Amosov A.M., Noshikhin D.V., Fomichev A.V., Galepin K.E., Savin A.N., Malinin D.S., Vasiliev V.V., Spirin V.V., Avdeenko V.V., Preobrazhenskaya A.A., and Matushkina G.L. for their technical discussions and professional insights.

Colleagues at the Financial University under the Government of the Russian Federation, including Dean of the Faculty of Information Technology Feklin V.G., Head of the Department of Big Data and Machine Learning Alyunov A.N., Petrosov D.A., Solovyov V.I., Zholobova G.N., Koroteev M.V., Ivanov M.N., Ivanyuk V.A., and Savinov E.A., are acknowledged for their contributions to the discussion of computational aspects of the work.

The author is also thankful for constructive discussions with research peers, including Kondrashkin I.B., Motorin N.M., Klepikov K.S., Mishchenko A.Yu., Kostenetsky A.L., Shevelev S.E., Isakov S.V., Klementyev A.V., Loginov V.Yu., Volkanov D.Yu., Foliyants E.V., Kondar E.V., Burova V.P., Vasilyeva O.A., Shabelnikov A.V., Serzhantova O.V., Danilova N.V., Khan T.A., Kuzanskii N.V., Khrushchev Yu.V., Egorkin I.A., and pulmonologist Krysin Yu.S. from the Central Clinical Hospital of the Presidential Administration of Russia for their feedback and insightful discussions.

Finally, the author acknowledges the broader scientific and professional community, whose constructive criticism and engagement contributed to the refinement of this work.

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1 Introduction

The present work is the result of an attempted reconstruction of Fermat's original discourse along with an explanation of why he might have not written it down. The author had performed it within a two-time period of time— between 1990 and 1993 – trying to prove the theorem and the final revision in 2017-2026.¹ When completed, it did look like a proof of Fermat's epoch, as it only involved the knowledge and techniques available and utilised by Fermat's contemporary and pre-Fermat mathematical world.

Not to overburden this text with details of a real historical study, let us briefly recall the history of the conjecture. Around 1637, Fermat wrote his Last Theorem in the margin of his copy of the *Arithmetica* next to Diophantus' sum-of-squares problem [7]:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadra-tos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

Attempts to prove this conjecture employed a diverse range of methods. Early attempts, including Fermat's own proof for the case $n = 4$, utilized the **method of infinite descent**. A significant step forward was Leonhard Euler's proof for the case $n = 3$, which involved the use of **Gaussian integers** [13]. Subsequent efforts included approaches based on the work of Sophie Germain and other techniques, before Andrew Wiles finally presented a complete proof in 1995, drawing upon deep results from **the theory of modular forms and elliptic curves** [1–6].

Modern methods, such as the theory of modular forms, deal with **the transformation properties of specific curves over particular types of spaces** (e.g., rational numbers), highlighting the **stability of elliptic curves with respect to modular transformations**. While the contemporary formalizations of algebraic curves, spaces, transformations, and groups used in these methods were not present in Fermat's time, mathematicians of that era possessed their own approaches and intuitive understandings for studying the properties of natural numbers (and primes). (More on this can be found in the Conclusions section of this paper.)

Theorem 1 (Conditional on the mutual scale symmetry of the two-power decomposition). *Assume that any hypothetical positive natural solution of the Fermat equation*

$$x^n + y^n = z^n$$

with $n > 2$ admits the real symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

with positive real parameters $m, p > 0$. Then

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

Let the odd binomial core of this difference be represented in two natural ways by positive real residual scales $l, q > 0$:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n,$$

¹ The work was mainly performed on the scholarship of the NRNU MEPhI student in 1990 – 93, final revision in 2017 – 2026.

and

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n.$$

Here l is the residual scale obtained after extracting the linear binomial factor n , while q is the residual scale obtained after extracting the binary odd-coefficient mass 2^{n-1} . In this theorem, q is a positive real scale and must not be confused with a modulus in modular arithmetic.

Assume further the mutual scale-symmetry condition:

$$l^n = q^n.$$

Then the Fermat equation

$$x^n + y^n = z^n$$

has no positive natural solutions for any $n > 2$.

Proof sketch. Given a putative positive natural solution at a fixed exponent $n > 2$, the two-power decomposition gives the binomial identity

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of the binomial expansion admits two real normalizations:

$$y^n = 2n l^n$$

and

$$y^n = 2^n q^n.$$

Equating these two expressions gives

$$2n l^n = 2^n q^n.$$

Since $2n > 0$, we can divide by $2n$ and obtain

$$l^n = q^n \frac{2^{n-1}}{n}.$$

By the mutual scale-symmetry condition,

$$l^n = q^n.$$

Substituting this into the previous relation yields

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, we have $q^n > 0$, and therefore cancellation of q^n is legitimate. Hence

$$1 = \frac{2^{n-1}}{n},$$

or equivalently

$$n = 2^{n-1}.$$

The equation

$$n = 2^{n-1}$$

has exactly two positive integer solutions:

$$n \in \{1, 2\}.$$

Indeed, it holds for $n = 1$ and $n = 2$, while for every integer $n > 2$ the exponential term 2^{n-1} strictly dominates the linear term n :

$$2^{n-1} > n.$$

Therefore a putative positive natural solution with $n > 2$ leads to a contradiction.

Thus, under the mutual scale-symmetry condition $l^n = q^n$, the only admissible exponents are $n = 1$ and $n = 2$, and no positive natural solution exists for any exponent higher than the square. The elementary growth comparison

$$n = 2^{n-1} \implies n \in \{1, 2\}$$

is the same final exponential-linear obstruction that is machine-checkable in the accompanying Coq file `GlobalNormalization.v`.

2 Possible Conditional Reconstruction

The statement of the theorem is rather straightforward and as follows:

Neither a cube for two cubes, nor a biquadrate or two biquadrates, and generally no power greater than two can be decomposed into two powers of the same grade. In other words, the equation

$$x^n + y^n = z^n$$

has no solutions in natural numbers if n is an integer greater than 2.

Therefore, first

- 1). Let's write down the theorem

$$x^n + y^n = z^n \tag{2.1}$$

- 2). Let's rewrite (2.1)

$$z^n - x^n = y^n \tag{2.2}$$

- 3). Let's perform the transformation of (2.2) to

$$(m^n + p^n)^n - (m^n - p^n)^n = y^n, \tag{2.3}$$

where $n \in \mathbb{N}$, m, p are arbitrary numbers (not necessarily integers; signs arbitrary), and $z, x \in \mathbb{N}$ satisfy

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{2.4}$$

Raising both identities to the n -th power yields

$$\begin{cases} (m^n + p^n)^n = z^n, \\ (m^n - p^n)^n = x^n. \end{cases} \tag{2.5}$$

Thus (2.4) implies (2.5).

Solving (2.4). Adding and subtracting the equations gives

$$2m^n = z + x, \quad 2p^n = z - x,$$

so formally

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}.$$

Domain note. Over $\mathbb{R}$: for odd n the roots are unique; for even n we need $(z \pm x)/2 \geq 0$ and there is a sign choice $m = \pm((z+x)/2)^{1/n}$, $p = \pm((z-x)/2)^{1/n}$. Over $\mathbb{C}$: there are n branches for the n -th root; once a branch is fixed, the reconstruction is consistent.

Equivalence of (2.4)–(2.5). **Forward** (2.4) $\Rightarrow$ (2.5). Immediate by raising to the n -th power.

Reverse (2.5) $\Rightarrow$ (2.4). Assume $z, x \in \mathbb{N}$ and

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Taking n -th roots in the same domain as $m^n \pm p^n$ (principal root over $\mathbb{R}_{\geq 0}$, or a fixed branch over $\mathbb{C}$) gives

$$z = m^n + p^n, \quad x = m^n - p^n,$$

i.e. (2.4). For even n one fixes consistent signs as above. Hence, (2.4) and (2.5) are equivalent in the chosen number domain once the root/branch convention is fixed.

Proposition 1 (Integer case: necessary & sufficient conditions). *If, in addition, one requires $m, p \in \mathbb{Z}$, then necessarily*

$$z \pm x \text{ are even,} \quad \frac{z+x}{2} = m^n, \quad \frac{z-x}{2} = p^n.$$

Conversely, if $z \pm x$ are even and both halves are perfect n -th powers in $\mathbb{Z}$, then the reconstructed m, p are integers (up to signs for even n).

Remark 1. *This separates the general real/complex reconstruction (no parity obstruction) from the integer reconstruction, where parity and perfect-power constraints are essential.*

Example (real domain). For $n = 2$, $m = 3$, $p = 2$:

$$z = 3^2 + 2^2 = 13, \quad x = 3^2 - 2^2 = 5,$$

hence

$$z^2 = 169, \quad x^2 = 25,$$

and reversing via $(z \pm x)/2$ returns $m = 3$, $p = 2$.

Counterexample (integer case). For $n = 3$, $z = 2$, $x = 1$ we have $(z+x)/2 = 3/2$, which is not a perfect cube in $\mathbb{Z}$; hence no integer m, p satisfy (2.4), although real m, p exist.

Modular remark. The present work does not construct a proof by means of modular arithmetic and does not claim the absence of solutions to the corresponding congruences over finite fields. The scheme considered here concerns equalities over the natural, integer, rational, and real numbers, not equations over a finite field $\mathbb{F}_q$.

In particular, from an integer equality

$$x^n + y^n = z^n$$

one indeed obtains the congruence modulo any q :

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse implication is not valid in general: from the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

one cannot infer the equality

$$x^n + y^n = z^n$$

over the integers.

In other words, a modular congruence only means that there exists an integer t such that

$$z^n = x^n + y^n + qt.$$

An integer solution of Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not disprove the real parametrization used in the present work.

If one specifically passes to a modulus q , then the parameters must be specified separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one should separately fix the residues

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q},$$

after which the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

Thus, the real parametrization and modular arithmetic belong to different levels of description. In this article, the first level is used: equalities over $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, rather than a separate theory of solutions over finite fields.

- 4). Let's ask ourselves the question: what is the number y ? Is it positive integer or not? If it is not positive integer, then under what conditions will it be natural number? Does its naturalness depend on the degree of n ?

From (2.3) we have the difference:

$$y^n = z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = \quad (2.6)$$

that can be expanded or decomposed into a sum according to Newton's binomial [8, 9]:

$$\begin{aligned} &= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \dots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] - \\ &- [(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \dots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^1(m^n)^{n-1}p^n + 2C_n^3(m^n)^{n-3}(p^n)^3 + \dots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2 \sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

Rewrite then (2.6) as

$$\begin{aligned} &z^n = x^n + y^n, \\ &\text{where } x, y, z \text{ are } \begin{cases} z = m^n + p^n \\ x = m^n - p^n \\ y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \right]^{1/n} \end{cases} \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even;

we know that in common case for $n = 2$ we have

$$\text{where } x, y, z \text{ are given by } \begin{cases} z = r \cdot (m^2 + p^2), \\ x = r \cdot (m^2 - p^2), \\ y = r \cdot 2mp. \end{cases} \quad (2.7)$$

but in our case, we omit the factor r – is some positive ($r > 0$) integer constant since it is reduced in our calculations. Therefore, we made the transformation (2.3) in order to take into account this case ($n=2$), known since the time of Pythagoras, since the general case should include a particular solution as a subset.

- 5). scrutinise now the y :

$$y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \right]^{1/n}. \quad (2.8)$$

In order for the y to maybe a positive integer, $\sqrt[n]{2}$ must leave, since for $n > 1$ $\sqrt[n]{2}$ is an irrational number (see Appendix A). It is thus necessary that the expression

$$\left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} \quad (2.9)$$

contain some common factor that destroys the radical expression $\sqrt[n]{2}$, let's find out what it is. Otherwise, y is not a positive integer due to the presence of $\sqrt[n]{2}$. Consider now what largest divisor this sum may contain and what it is equal to:

$$\begin{aligned} & \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right] = \\ &= C_n^1 (m^n)^{n-1} p^n + C_n^3 (m^n)^{n-3} (p^n)^3 + \dots + C_n^k (m^n)^{n-k} (p^n)^k + \{C_n^n (p^n)^n\} = \\ &= n(m^n)^{n-1} p^n + \frac{n(n-1)(n-2)}{3!} (m^n)^{n-3} p^3 + \dots + \\ &+ \frac{n(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{n(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} = \\ &= n \cdot \left[(m^n)^{n-1} p^n + \frac{(n-1)(n-2)}{3!} (m^n)^{n-3} p^3 + \dots + \right. \\ &\left. + \frac{(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} \right] = n \cdot l^n \end{aligned} \quad (2.10)$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even,

Hence the conclusion follows: the expression admits a common factor n in the real algebraic factorization. Here l denotes a scaling constant about which nothing is assumed a priori; in the general case it may be real, and in special arithmetic cases it may be integer. It represents the remaining part of the radical after extracting the common factor n .

Since each term in the expression can be written with a common factor n , the whole sum can be factored over the real domain in the form

$$\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1} = n l^n.$$

Here l is a real scaling parameter defined by this factorization. Thus, this step should be understood as extracting a common real factor n , not as claiming integer divisibility of the whole expression by n .

To show the uniqueness of the decomposition of expression (2.10), we can use the following considerations:

- (a) Degree n is a fixed positive integer.
- (b) Exponentiation is an unambiguous operation; for any number, there is a unique value for the degree.
- (c) Addition and subtraction of real numbers are commutative operations, and the result does not depend on the order of actions.

Based on these properties, it can be argued that for fixed m , p and n there is a single result of calculating the expression (2.10). Rearranging the members will not affect the final answer.

We can see from (2.8 - 2.10) (where l is some constant (which will be it for fixed m , p , n), which we don't know anything about yet)

$$y = \sqrt[n]{2 \cdot n \cdot l} \quad (2.11)$$

As a result, from (2.6)–(2.11) we get

$$z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = y^n. \quad (2.12)$$

Substituting (2.11) into (2.12), we have

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \cdot n l^n. \quad (2.13)$$

The second symmetric scale and the mutual-scale condition.

Let us denote the odd binomial core by

$$S_n(m, p) := \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1},$$

where $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even. Then the binomial expansion gives

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p).$$

From the previous real factorization of the odd binomial core we have

$$S_n(m, p) = n l^n,$$

where $l > 0$ is a real scaling parameter. Hence

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n. \quad (2.13)$$

On the other hand, the sum of the odd binomial coefficients satisfies the classical identity

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1}.$$

This identity expresses the binary symmetry of Pascal's triangle: the total mass of the odd coefficients equals the total mass of the even coefficients and is equal to one half of the full binomial mass 2^n .

Therefore, we may introduce a second real scale $q > 0$, associated with this coefficient-normalized binary decomposition, by writing

$$S_n(m, p) = 2^{n-1} q^n.$$

Equivalently,

$$q^n = \frac{1}{2^{n-1}} \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1}.$$

This is not a claim that the weighted binomial core is equal to the pure sum of its coefficients. Rather, it is the definition of the second real residual scale q after the coefficient mass 2^{n-1} has been extracted.

Thus the same difference of powers also admits the representation

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n. \quad (2.14)$$

Comparing (2.13) and (2.14), we obtain

$$2n l^n = 2^n q^n.$$

After division by $2n > 0$, this gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (2.15)$$

Mutual symmetry of the two scales. The two quantities l and q arise from two natural normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

The first normalization extracts the linear factor n , which is present in the odd binomial coefficients. The second normalization extracts the total odd binomial mass 2^{n-1} , which comes from the binary symmetry of the coefficients.

We shall call the following condition the mutual scale-symmetry condition:

$$l^n = q^n. \tag{2.16}$$

This condition states that the two natural residual scales of the same symmetric two-power decomposition coincide.

Combining (2.15) with (2.16), we get

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, we have $q^n > 0$, and therefore cancellation of q^n is legitimate. Consequently,

$$1 = \frac{2^{n-1}}{n}.$$

Equivalently,

$$n = 2^{n-1}. \tag{2.17}$$

Analysis of the equation $n = 2^{n-1}$.

The equation (2.17) is equivalent to

$$2^n = 2n.$$

It has exactly two positive integer solutions:

$$n = 1, \quad n = 2.$$

Indeed,

$$2^{1-1} = 1, \quad 2^{2-1} = 2.$$

For every integer $n > 2$, the exponential term 2^{n-1} strictly dominates the linear term n . Hence

$$2^{n-1} > n \quad \text{for all } n > 2.$$

Therefore, equation (2.17) is impossible for every exponent higher than the square.

Conclusion under mutual scale symmetry. If the mutual scale-symmetry condition

$$l^n = q^n$$

is a consequence of the intrinsic binary symmetry of every hypothetical positive integer solution of

$$x^n + y^n = z^n,$$

then the equality

$$n = 2^{n-1}$$

follows for every such solution. But this equality is possible only for

$$n \in \{1, 2\}.$$

Consequently, no positive integer solution can exist for any $n > 2$.

Thus, the decisive point is the status of the symmetry condition

$$l^n = q^n.$$

If it is proved from the two-power symmetry of the Fermat equation itself, then the argument becomes unconditional at this point. If it is assumed as an additional normalization principle, then the result remains a conditional theorem under the mutual scale-symmetry hypothesis.

6.) Verification (without considering the total multiplier r from (2.7), which has been reduced)

(a) Consider the case $n = 1$

$$\begin{aligned} z &= m + p \\ x &= m - p \\ y &= 2 \cdot [C_1^1(m^1)^{1-(2 \cdot 0+1)}(p^1)^{(2 \cdot 0+1)}] = 2[1 \cdot 1 \cdot p] = 2p \end{aligned}$$

that is, for the case $n = 1$, we have a solution in natural numbers x, y, z , if m and p are positive integer numbers and $m > p$.

(b) Consider the case $n = 2$

$$\begin{aligned} z &= m^2 + p^2 \\ x &= m^2 - p^2 \\ y &= \sqrt{2} \cdot [C_1^2(m^2)^{2-(2 \cdot 0+1)}(p^2)^{(2 \cdot 0+1)}]^{1/2} = \sqrt{2} \cdot [2m^2p^2]^{1/2} = 2mp \end{aligned}$$

that is, for the case $n = 2$, we also have a solution in positive integer numbers x, y, z , if m and p are positive integer numbers and $m > p$, or according to [10], if $m > p$, and m, p are such non-integer numbers that combinations with them will give the positive integer numbers x, y, z , for example: $m = 3/\sqrt{2}$, $p = 1/\sqrt{2}$, in this case, we also get the classical Pythagorean triple:

$$\begin{aligned} x &= m^2 - p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} - \frac{1}{2} = \frac{8}{2} = 4 \\ y &= 2 \cdot m \cdot p = 2 \cdot \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 3 \\ z &= m^2 + p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} + \frac{1}{2} = 5 \end{aligned}$$

We have obtained the classical Pythagorean triple and can see that this formula works.

7). Let us now consider the more general case where r is not reduced to proceed to final formulas like (2.7). If r is a positive integer number, then all the above applies. But according to [10], r can be a rational number. For example, let's say $r = 0.5$:

(a) Consider the case of $n = 1$, $m = 4$, and $p = 2$

$$\begin{aligned} x &= 0.5 \cdot (m - p) = 0.5 \cdot (4 - 2) = 1 \\ z &= 0.5 \cdot (m + p) = 0.5 \cdot (4 + 2) = 3 \\ y &= 0.5 \cdot (2 \cdot p) = 0.5 \cdot (2 \cdot 2) = 2 \end{aligned}$$

We see positive integer numbers. That is, m, p must be such that 7(a) is performed similarly. In this case, we can see that the formula 6(a) works.

(b) Consider the case of $n = 2$, $m = 2 \cdot \sqrt{2}$, $p = \sqrt{2}$,

$$\begin{aligned} x &= 0.5 \cdot (m^2 - p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 - (\sqrt{2})^2\right) = 3 \\ z &= 0.5 \cdot (m^2 + p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 + (\sqrt{2})^2\right) = 5 \\ y &= 0.5 \cdot (2 \cdot m \cdot p) = 0.5 \cdot (2 \cdot (2 \cdot \sqrt{2}) \cdot (\sqrt{2})) = 4 \end{aligned}$$

We see positive integer numbers. That is, m, p must be such that 7(b) holds similarly. In this case, we can see that the formula 6(b) works.

This preserves the generality of solutions for $n = 2$, as seen in the Pythagorean triple example.

8). Therefore, the cases $n = 1$ and $n = 2$ **exhaust all possible integer solutions**, which is consistent with the theorem.

The test showed that for $n=1$ or for $n=2$, in all the cases considered, we have solutions of the equation $x^n + y^n = z^n$ in the positive integer numbers x, y, z

9). **the equation $x^n + y^n = z^n$ has roots in the positive integer numbers x, y, z only for $n = 1$ and for $n = 2$**

**This completes the conditional argument
under the mutual scale-symmetry hypothesis.**

3 Remark and corollaries

REMARK. Note that the expression (2.8) can be simplified, namely

$$\begin{aligned}
 y &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} = \\
 &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \frac{(m^n)^n}{(m^n)^{2i} m^n} (p^n)^{2i} p^n \right]^{1/n} = \\
 &= \sqrt[n]{2} m^n \frac{1}{m} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{n2i} \right]^{1/n} = \\
 &= \sqrt[n]{2} m^{n-1} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{2in} \right]^{1/n}
 \end{aligned} \tag{3.1}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

COROLLARY 1. Consider the case $m = p$, then from the expression (3.1) it can be derived that

$$\begin{aligned}
 x &= 0 \\
 z &= m^n + m^n = 2m^n \\
 y &= \sqrt[n]{2} m^{n-1} m \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{m}{m} \right)^{2in} \right]^{1/n} = \sqrt[n]{2} m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n}
 \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

It is obvious that $y = z$

$$\begin{aligned}
 \sqrt[n]{2} m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} &= 2m^n \\
 \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} &= 2
 \end{aligned} ,$$

whence

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1} \tag{3.2}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

COROLLARY 2. Based on (3.2), the sum of even combinations can be calculated.

Consider Pascal's triangle [7]:

$$\begin{array}{ccccccc}
 & & 1 & & 1 & & \\
 & & 1 & & 2 & & 1 \\
 & & 1 & & 3 & & 3 & & 1 \\
 & & 1 & & 4 & & 6 & & 4 & & 1 \\
 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

Similarly to the above, it is concluded that

$$\sum_{j=0}^s C_n^{2j} = 2^{n-1} \tag{3.3}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

PROOF.

Expand

$$(m^n + p^n)^n + (m^n - p^n)^n =$$

into a binomial [8, 9]:

$$\begin{aligned}
 &= [(m^n)^n + C_n^1 (m^n)^{n-1} p^n + C_n^2 (m^n)^{n-2} (p^n)^2 + \dots + C_n^{n-1} m^n (p^n)^{n-1} + (p^n)^n] + \\
 &+ [(m^n)^n - C_n^1 (m^n)^{n-1} p^n + C_n^2 (m^n)^{n-2} (p^n)^2 \pm \dots \pm C_n^{n-1} m^n (p^n)^{n-1} \pm (p^n)^n] = \\
 &= 2C_n^0 (m^n)^n + 2C_n^2 (m^n)^{n-2} (p^n)^2 + \dots + 2C_n^k (m^n)^{n-k} (p^n)^k + \{2C_n^n (p^n)^n\} = \\
 &= 2 \sum_{j=0}^s C_n^{2j} (m^n)^{n-2j} (p^n)^{2j}
 \end{aligned}$$

with $s = (n-1)/2$ if n is odd and $s = n/2$ if n is even.

If $m = p$

$$(p^n + p^n)^n + (p^n - p^n)^n = 2 \sum_{j=0}^s C_n^{2j} (p^n)^{n-2j} (p^n)^{2j}$$

$$(2p^n)^n = 2 \sum_{j=0}^s C_n^{2j} \frac{(p^n)^n}{(p^n)^{2j}} (p^n)^{2j}$$

$$2^n (p^n)^n = 2 (p^n)^n \sum_{j=0}^s C_n^{2j}$$

$$2^{n-1} = \sum_{j=0}^s C_n^{2j}$$

with $s = (n - 1)/2$ if n is odd and $s = n/2$ if n is even.

Corollary 2 is proved.

COROLLARY 3. *Analysing (3.2) and (3.3), it can be concluded that*

$$\sum_{i=0}^k C_n^{(2i+1)} = \sum_{j=0}^s C_n^{2j} \quad (3.4)$$

with $k = (n - 1)/2$, $s = (n - 1)/2$ if n is odd and $k = (n - 2)/2$, $s = n/2$ if n is even.

Why such borders?

Type of number	Condition	Last constant in equation
Odd	$k = 2((n-1)/2)+1 = n$	C_n^n
	$s = 2((n-1)/2) = n-1$	C_n^{n-1}
Even	$k = 2((n-2)/2)+1 = n-1$	C_n^{n-1}
	$s = 2(n/2) = n$	C_n^n

Table 1: Boundaries of binomial coefficients

Conclusion: the sum of even coefficients is equal to the sum of odd ones and is equal to 2^{n-1} , therefore from (3.4) we have

$$\sum_{r=0}^n C_n^r = \sum_{i=0}^k C_n^{(2i+1)} + \sum_{j=0}^s C_n^{2j} = 2 \cdot 2^{n-1} = 2^n \quad (3.5)$$

with $k = (n - 1)/2$, $s = (n - 1)/2$ if n is odd and $k = (n - 2)/2$, $s = n/2$ if n is even.

4 Conclusion

The “difficulties” were for Fermat the lengthiness of the run of his deductions *put in writing*, as in the first half of the seventeenth century the mathematical notations had been way far from their present concise and diverse shape, many actions had to be written down *in words*. **Besides, a purely mathematical challenge was that he had to operate the then entirely new notions of binomials and logarithms**, both having just appeared for use and to be learnt “on the fly”.

As mentioned in the introduction, the mathematical methods from Pierre Fermat’s era used in this conditional reconstruction are accessible to any first-year physics and mathematics student. This contrasts favorably with Andrew Wiles’s proof, which is quite complex for the average mathematician due to its use of advanced and intricate modern mathematical tools.

Fermat was, plausibly, “playing” with the new algebraic ideas of his time, decomposing powers of sums and differences and observing the special role of the binary structure. In the reconstruction

proposed here, this leads not directly to an unconditional conclusion, but to a conditional scale relation: once the two natural residual scales of the odd binomial core are required to coincide,

$$l^n = q^n,$$

the comparison of the two normalizations

$$y^n = 2n l^n, \quad y^n = 2^n q^n$$

immediately yields

$$n = 2^{n-1}.$$

This elementary exponential-linear equation has positive integer solutions only for $n = 1$ and $n = 2$. Thus, under the mutual scale-symmetry condition, the equation

$$x^n + y^n = z^n$$

can have positive integer solutions only in the linear and quadratic cases.

He (would have) had first to introduce the two new notions so as to fully explain his finding. One can imagine *how much room it would take to put down all the deliberations* that had led him to his discovery *on the margins of a book solely without the proper symbolic notations that a contemporary mathematician avails*.

Why Pierre Fermat did not write down all those ideas in a dedicated document is the dedicated question of a dedicated research endeavour. It can come out that he had authored such a separate document indeed, which afterwards was somehow lost or – alternatively – has survived to this day, hidden in an archive or a library or in somebody's unrealised custody.

The author requests the mathematical society to look critically at the deliberations set forth above and to return their assessment.

This paper has been published as a preprint on the ResearchGate platform and on the OSF platform at the following links [11, 12].

The Coq formalization is available in the repository FLT-Coq on GitHub [14].

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A Appendix A. Proof of the Irrationality of $\sqrt[n]{2}$ for $n \geq 2$

A.1 Theorem: The number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

A.2 Proof:

We will prove by contradiction. Assume that $\sqrt[n]{2}$ is a rational number. This means it can be represented as an irreducible fraction $\frac{p}{q}$, where p and q are integers, $q \neq 0$, and p and q have no common divisors other than 1.

Thus, we have:

$$\sqrt[n]{2} = \frac{p}{q}$$

Raise both sides of the equation to the power of n :

$$(\sqrt[n]{2})^n = \left(\frac{p}{q}\right)^n$$

$$2 = \frac{p^n}{q^n}$$

Multiply both sides by q^n :

$$2q^n = p^n$$

From this equation, we see that p^n is an even number, since it is equal to $2q^n$. If p^n is even, then p must also be even (if p were odd, then p^n would also be odd).

Since p is even, we can write it as $p = 2k$, where k is some integer. Substitute this expression for p into the equation $2q^n = p^n$:

$$2q^n = (2k)^n$$

$$2q^n = 2^n k^n$$

Divide both sides of the equation by 2 (which is permissible since $n \geq 2$, and therefore 2^n is divisible by 2):

$$q^n = 2^{n-1} k^n$$

Since $n \geq 2$, then $n - 1 \geq 1$. From the last equation, we see that q^n is an even number, since it is equal to $2^{n-1} k^n$, where 2^{n-1} is an even factor. If q^n is even, then q must also be even.

Thus, we have reached the conclusion that both p and q are even numbers. This means they have a common divisor of 2. However, at the beginning of the proof, we assumed that the fraction $\frac{p}{q}$ was irreducible, meaning p and q have no common divisors other than 1.

The resulting contradiction indicates that our initial assumption about the rationality of $\sqrt[n]{2}$ is incorrect.

A.3 Conclusion:

Therefore, the number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

B Appendix B. Proof that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ is 1

B.1 Problem Statement

Prove that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

B.2 Solution

Consider the function $f(n) = \sqrt[n]{2 \cdot n}$. Our goal is to find the limit of this function as $n \rightarrow \infty$. Let's rewrite the function in the form of a power:

$$f(n) = (2 \cdot n)^{\frac{1}{n}}$$

To find the limit of this function, let's consider the natural logarithm of $f(n)$:

$$\ln(f(n)) = \ln\left((2 \cdot n)^{\frac{1}{n}}\right)$$

Using the properties of logarithms, we get:

$$\ln(f(n)) = \frac{1}{n} \ln(2 \cdot n)$$

Separate the logarithm of the product into the sum of logarithms:

$$\ln(f(n)) = \frac{\ln(2) + \ln(n)}{n}$$

Now, let's find the limit of $\ln(f(n))$ as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = \lim_{n \rightarrow \infty} \frac{\ln(2) + \ln(n)}{n}$$

This limit has the indeterminate form $\frac{\infty}{\infty}$, so we can apply L'Hôpital's Rule. Take the derivatives of the numerator and denominator with respect to n :

$$\frac{d}{dn}(\ln(2) + \ln(n)) = \frac{1}{n}$$

$$\frac{d}{dn}(n) = 1$$

Thus, the limit becomes:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

So, we have found that the limit of the natural logarithm of the function is 0:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = 0$$

Now, to find the limit of the original function $f(n)$, we use the continuity of the exponential function:

$$\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} e^{\ln(f(n))} = e^{\lim_{n \rightarrow \infty} \ln(f(n))} = e^0 = 1$$

B.3 Conclusion

Therefore, we have proven that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

C Appendix C. A short proof of monotonicity

It is well known that $\sqrt[n]{2 \cdot n} \rightarrow 1$ as $n \rightarrow \infty$ (see Appendix B). However, in this appendix we focus on showing that $(2 \cdot n)^{1/n}$ is strictly decreasing for $n \geq 3$. Below is one way to prove this, using the logarithmic derivative:

C.1 Define the function.

Let

$$f(n) = (2 \cdot n)^{\frac{1}{n}}.$$

It is convenient to work with its natural logarithm:

$$\ln(f(n)) = \ln((2 \cdot n)^{\frac{1}{n}}) = \frac{1}{n} \ln(2 \cdot n).$$

Denote

$$g(n) = \ln(f(n)) = \frac{\ln(2 \cdot n)}{n} = \frac{\ln(2) + \ln(n)}{n}.$$

C.2 Compute the derivative.

Treating n as a real variable $n > 0$, we have

$$g'(n) = \frac{d}{dn} \left(\frac{\ln(2 \cdot n)}{n} \right).$$

Using the quotient rule,

$$g'(n) = \frac{\frac{d}{dn}[\ln(2 \cdot n)] \cdot n - \ln(2 \cdot n) \cdot 1}{n^2} = \frac{\frac{1}{n} \cdot n - \ln(2 \cdot n)}{n^2} = \frac{1 - \ln(2 \cdot n)}{n^2}.$$

(Note that $\frac{d}{dn}[\ln(2 \cdot n)] = \frac{1}{2 \cdot n} \cdot 2 = \frac{1}{n}$.)

C.3 Sign of the derivative.

We want to see where $g'(n) < 0$:

$$g'(n) < 0 \iff 1 - \ln(2 \cdot n) < 0 \iff \ln(2 \cdot n) > 1 \iff 2 \cdot n > e \iff n > \frac{e}{2}.$$

Since $e \approx 2.718$, the inequality $n > e/2$ is certainly true for all integer $n \geq 2$, and hence strictly for $n \geq 3$. Therefore, for $n \geq 3$, $g(n)$ is strictly decreasing.

C.4 Implication for $f(n)$.

Since $f(n) = \exp(g(n))$, and $\exp(\cdot)$ is a strictly increasing function in its argument, $f(n)$ decreases whenever $g(n)$ decreases. Hence, $f(n)$ is indeed strictly decreasing for $n \geq 3$.

Thus, only for $n = 1$ and $n = 2$ does $f(n)$ attain the maximum value 2, and for all $n \geq 3$, it strictly decreases (which is consistent with the limit $\lim_{n \rightarrow \infty} (2 \cdot n)^{1/n} = 1$).

C.5 Conclusion

Hence, we have demonstrated that $f(n)$ is strictly decreasing for $n \geq 3$, either by analyzing the derivative of $\ln(f(n))$ (as shown above) or by comparing $f(n+1)$ and $f(n)$. This completes the proof of the monotonicity of $f(n)$.

D Appendix D: Analysis of Functions and Heuristic Visualization of Critical Points

D.1 Introduction

The goal of this analysis is to study the behavior of the auxiliary continuous function

$$f(p, q) = q - \sqrt[p]{qp},$$

and to identify its critical points, in particular the points where the second derivative $f''(p)$ equals zero.

This study is not intended as an independent proof of Fermat's Last Theorem. Rather, it provides a geometric and analytic visualization of the special role of the binary value $q = 2$ in relation to the conditional scale-symmetry scheme discussed in the main article:

$$x^n + y^n = z^n.$$

In the main text, the decisive algebraic step is expressed in terms of two positive real residual scales l and q , arising from two natural normalizations of the same odd binomial core:

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Comparing these two representations gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

The conditional step is the mutual scale-symmetry condition

$$l^n = q^n.$$

Under this condition one obtains

$$n = 2^{n-1},$$

which has positive integer solutions only for $n = 1$ and $n = 2$.

In this appendix, the symbol q denotes an auxiliary positive real parameter in the function $f(p, q)$. It should not be confused with a modulus q used in modular arithmetic or with the finite field notation $\mathbb{F}_q$. It should also be distinguished from the residual scale q in the main algebraic text, although the same letter is used because both quantities are intended to visualize a positive scaling value. The auxiliary value $q = 2$ is interpreted here only as a geometric marker of the binary/quadratic case, not as an additional axiom and not as a replacement for the mutual scale-symmetry condition.

Thus, the role of this appendix is heuristic and illustrative. It visualizes why the point $p = 2$, $q = 2$ is geometrically distinguished, while the rigorous conditional obstruction in the article remains the algebraic implication

$$l^n = q^n \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

D.2 Results of the Analysis

D.2.1 Combined Graph of Functions

The following graph (Fig. 1) illustrates three functions, each with its corresponding discrete values:

- $1.5 - \sqrt[3]{1.5p}$ (blue line) and $1.5 - \sqrt[3]{1.5n}$ (blue squares),
- $2 - \sqrt[3]{2p}$ (green line) and $2 - \sqrt[3]{2n}$ (green squares),
- $3 - \sqrt[3]{3p}$ (orange line) and $3 - \sqrt[3]{3n}$ (orange squares).

These curves should be read as a visual comparison between continuous values of p and the discrete exponent values n . The case $q = 2$ is singled out because it corresponds to the binary value appearing in the equivalent exponential-linear relation

$$n = 2^{n-1} \quad \Longleftrightarrow \quad 2^n = 2n.$$

In the main text, this relation is not introduced by the graph itself. It follows from the comparison of the two binomial normalizations together with the mutual scale-symmetry condition $l^n = q^n$.

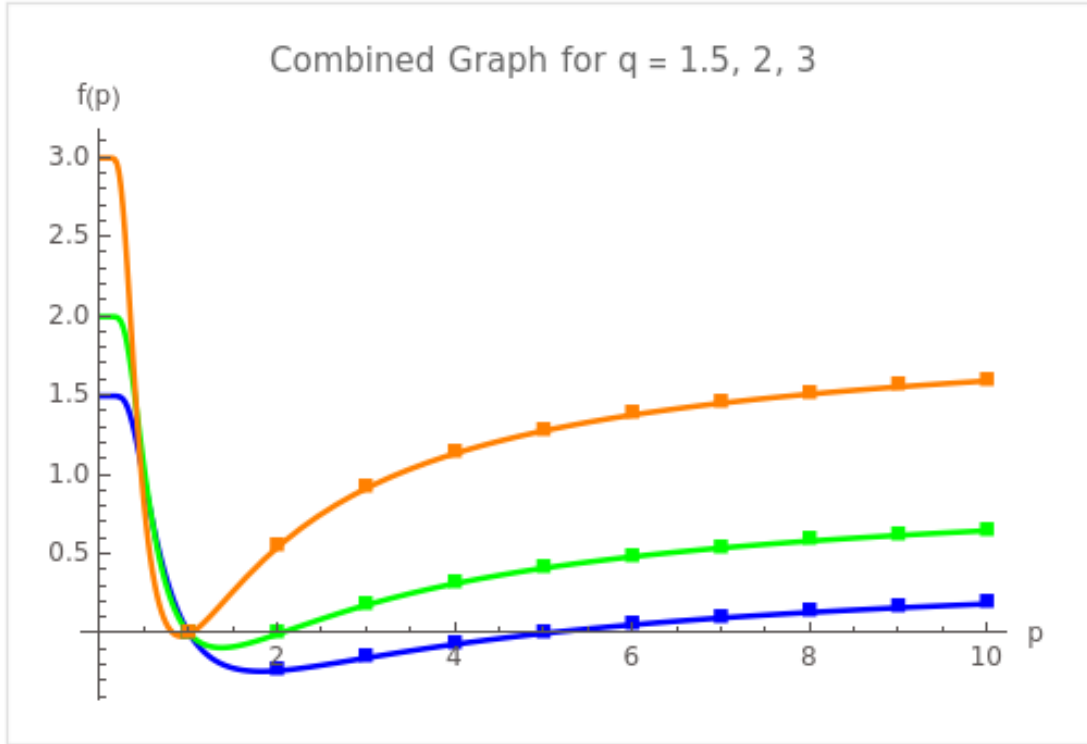


Figure 1: Combined graph of $q - \sqrt[3]{qp}$ for $q = 1.5, 2, 3$. Lines represent continuous p , and squares represent discrete n . The value $q = 2$ marks the binary/quadratic alignment.

D.2.2 Surface Plot of $f(p, q)$

The three-dimensional surface plot below (Fig. 2) illustrates the behavior of

$$f(p, q) = q - \sqrt[p]{qp}$$

for continuous values of $p > 0$ and $q > 0$. The perspective highlights the change in curvature as both parameters vary.

This visualization is meant to support the geometric intuition behind the binary value $q = 2$. The algebraic argument itself rests on the conditional comparison of the two residual scales l and q :

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

and on the mutual scale-symmetry condition

$$l^n = q^n.$$

Therefore, the surface plot is only a heuristic visual aid. It does not prove the scale-symmetry condition and does not replace the algebraic reduction to

$$n = 2^{n-1}.$$

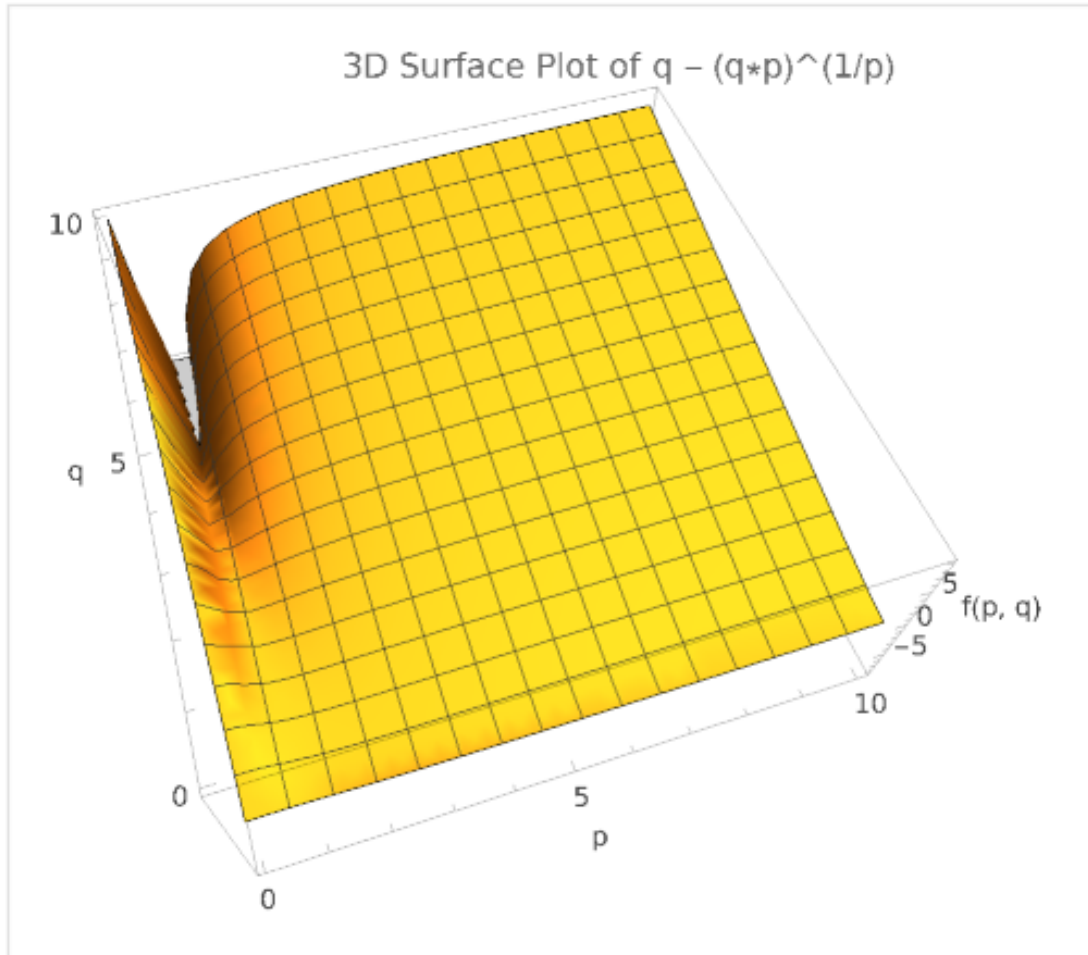


Figure 2: 3D surface plot of $f(p, q) = q - \sqrt[p]{qp}$ for $p > 0$ and $q > 0$. The point $p = 2$, $q = 2$ is interpreted as a geometric marker of the quadratic alignment.

D.3 Analysis and Conclusions

The numerical and graphical analysis suggests the following interpretation:

- The critical point at $p = 2$ appears naturally for $q = 2$, where the second derivative $f''(p)$ equals zero. This highlights the special geometric role of $q = 2$, corresponding to the binary/quadratic alignment.
- For other values of q , critical points may exist, but they occur at values $p \neq 2$. This illustrates that $q = 2$ is distinguished by its alignment with the quadratic case $n = 2$, which is precisely the case of Pythagorean triples.
- In the terminology of the main text, this does not mean that a scaling value is imposed by the graph. Rather, it gives a geometric picture compatible with the algebraic fact that the equation

$$n = 2^{n-1}$$

admits precisely the two positive integer solutions $n = 1$ and $n = 2$.

These results should be understood as a geometric consistency check for the algebraic argument of the article. They illustrate the same special role of the binary value

$$2,$$

but they do not constitute an independent proof of Fermat's Last Theorem. The rigorous conditional part of the argument remains the algebraic reduction

$$l^n = q^n \implies n = 2^{n-1},$$

together with the elementary, and machine-checkable, fact that this equation has only the positive integer solutions

$$n \in \{1, 2\}.$$

Appendix E: Geometric Motivation for the Mutual Scale-Symmetry Condition through the Function $f(p, q)$

In this appendix, we further investigate the connection between the auxiliary function

$$f(p, q) = q - \sqrt[n]{qp}$$

and the scale comparison associated with Fermat's equation

$$x^n + y^n = z^n.$$

The purpose of this appendix is not to provide a standalone geometric proof of Fermat's Last Theorem, but to visualize why the binary/quadratic point

$$p = 2, \quad q = 2$$

is geometrically distinguished.

In the conditional scheme of the main article, the decisive algebraic step is not the graph itself. The decisive step is the comparison of two real normalizations of the same odd binomial core:

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Their comparison gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If the two residual scales are required to coincide,

$$l^n = q^n,$$

then

$$1 = \frac{2^{n-1}}{n},$$

or equivalently

$$n = 2^{n-1}.$$

Thus, the final obstruction is the elementary exponential-linear equation

$$n = 2^{n-1},$$

not a graphical property of $f(p, q)$.

The function $f(p, q)$ gives a visual way to see why the point $p = 2, q = 2$ plays a special role: it marks the place where the binary value aligns with the quadratic case, that is, with the existence of Pythagorean triples. In this sense the function supports the geometric intuition behind the mutual scale-symmetry condition, but it does not prove that condition.

E.1 Analysis of the Second Derivative

The function

$$f(p, q) = q - \sqrt[n]{qp}$$

allows us to study the behavior of inflection points, which are determined by the condition

$$f_p''(p, q) = \frac{\partial^2}{\partial p^2} \left(q - \sqrt[n]{qp} \right) = 0.$$

Inflection points are useful here because they reveal changes in the shape of the auxiliary function. In the present context, these changes are interpreted as a geometric reflection of the transition from the quadratic case $n = 2$ to higher exponents $n > 2$.

The notation p in this appendix is used as a continuous variable, comparable with the discrete exponent n . It should not be confused with the parameter p in the main parametrization

$$z = m^n + p^n, \quad x = m^n - p^n.$$

Likewise, the notation q in this appendix is a positive real visualization parameter; it is not a modulus and is not a finite-field parameter.

E.2 Additional Analysis for the Variable q

Investigations were also conducted on the variable q . At the point $q = 2$, with $p = 2$, both the first and second partial derivatives with respect to q are equal to 0.5. This result indicates a special local behavior of the auxiliary function at the binary/quadratic point.

However, in the main argument of the article, the value $q = 2$ is not introduced as a consequence of this numerical observation. Nor is the mutual scale-symmetry condition

$$l^n = q^n$$

derived from the graph. Rather, the graph should be read only as an additional geometric illustration of why the quadratic alignment $p = 2, q = 2$ is visually distinguished.

Thus, the differential analysis in this appendix should be read as an auxiliary geometric commentary, not as a replacement for the algebraic scale comparison

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

nor as a proof of the condition

$$l^n = q^n.$$

E.3 Numerical Results

Below are the numerical values of the second derivative $f_p''(p, q)$ for various values of p and q :

p	$q = 1.5$	$q = 2.0$	$q = 2.5$	$q = 3.0$	$q = 4.0$
1	2.75	0.77	2.90	0.96	1.10
2	0.17	0.00	-0.11	-0.34	-0.55
3	-0.0056	-0.08	-0.099	-0.21	-0.31
4	-0.018	-0.11	-0.056	-0.12	-0.17
5	-0.014	-0.10	-0.033	-0.067	-0.093

Table 2: Numerical values of the second derivative $f_p''(p, q)$ for various p and q .

The table shows that the pair

$$p = 2, \quad q = 2$$

is singled out by the vanishing of the second derivative in the numerical data. This is consistent with the special role of the quadratic case, but it should not be interpreted as a direct proof that higher-degree integer solutions are impossible.

It is important to emphasize that this table is not used to derive the equality

$$n = 2^{n-1}.$$

The equality

$$n = 2^{n-1}$$

comes from the algebraic comparison of the two normalizations under the mutual scale-symmetry condition. The numerical table merely visualizes why the point associated with the quadratic case is geometrically distinguished.

E.4 Conclusions from the Data

- For $p = 2, q = 2$ we have $f_p''(p, q) = 0$, which geometrically corresponds to the quadratic case and is consistent with the existence of Pythagorean triples.
- For $q > 2$, the inflection point shifts toward lower values of p . This suggests a change in the shape of the auxiliary function and is consistent with the loss of the special quadratic alignment.
- Figure 3 illustrates the surface of $f_p''(p, q)$ with the marked inflection points.
- This geometric observation supports the interpretation that the binary/quadratic point $p = 2, q = 2$ is special, but it does not by itself prove the mutual scale-symmetry condition $l^n = q^n$.

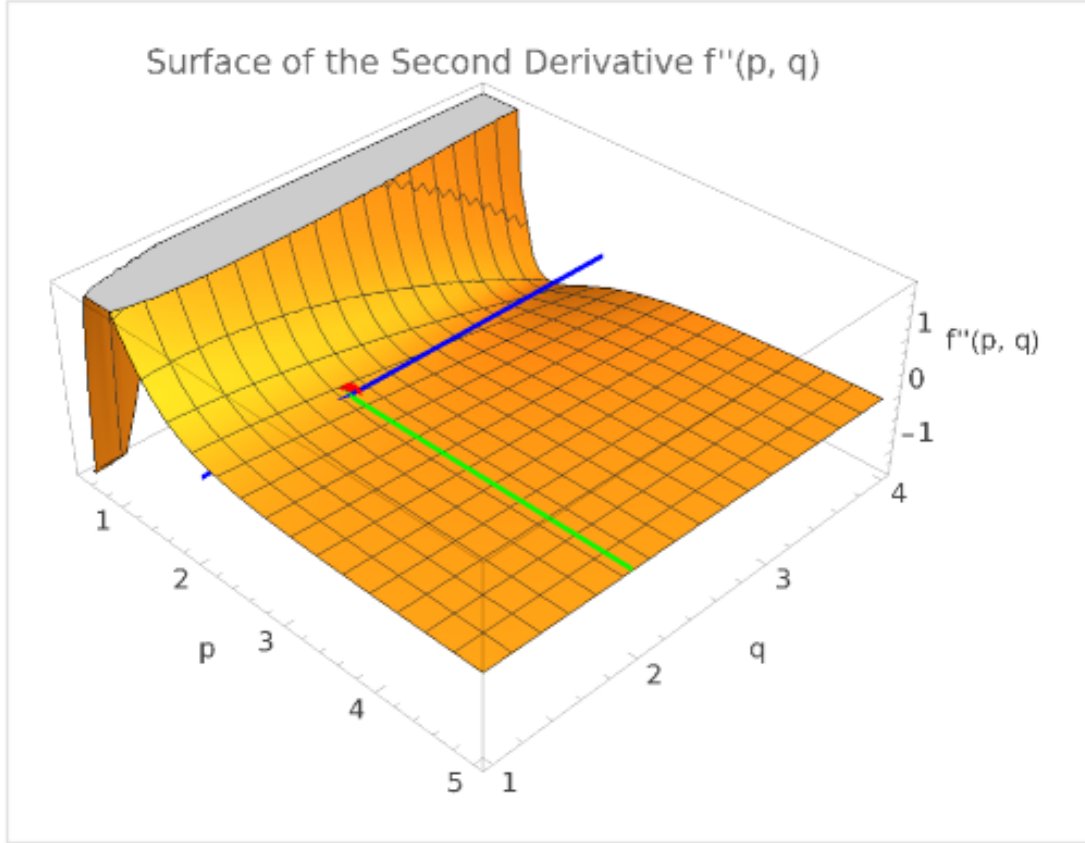


Figure 3: Surface of the second derivative $f_p''(p, q)$, demonstrating the shift of the inflection point and the special role of the point $p = 2, q = 2$.

E.5 Geometric Interpretation in Relation to Fermat's Equation

E.5.1 Relation to the Inflection Points

From the data, one can draw the following heuristic interpretation:

1. If the inflection point is aligned with the discrete exponent value $p = n$, then the auxiliary geometry is compatible with the corresponding exponent.
2. The alignment $p = 2, q = 2$ is consistent with the quadratic case $n = 2$, where integer solutions exist in the form of Pythagorean triples.
3. For $q > 2$, the shift of the inflection point away from $p = 2$ illustrates the loss of this quadratic alignment.
4. This loss of geometric alignment is compatible with, but does not prove, the algebraic obstruction obtained from the mutual scale-symmetry condition.

Thus, the geometric picture distinguishes the quadratic case

$$n = 2,$$

while the exclusion of exponents

$$n > 2$$

comes from the conditional algebraic relation

$$l^n = q^n \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

E.5.2 Final Conclusion

Based on the numerical and graphical calculations:

- **For $q = 2$, the inflection point corresponds to $p = 2 \Rightarrow$** the auxiliary geometry is aligned with the quadratic case, where Fermat's equation has integer solutions through Pythagorean triples.
- **For $q > 2$, the inflection point shifts to lower values of $p \Rightarrow$** the auxiliary geometry no longer has the same alignment with the quadratic case.
- This supports the interpretation that the binary/quadratic point

$$p = 2, \quad q = 2$$

is geometrically distinguished in the conditional framework of the article.

Thus, the function $f(p, q)$ provides a useful geometric visualization of the special role of the binary value $q = 2$ and of the quadratic exponent $p = 2$. It supports and illustrates the main conditional algebraic argument, but the strict exclusion of positive integer solutions for

$$x^n + y^n = z^n, \quad n > 2,$$

rests on the algebraic comparison

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

together with the separately stated mutual scale-symmetry condition

$$l^n = q^n.$$

Under this condition the relation

$$n = 2^{n-1}$$

follows, and this equation has only the positive integer solutions

$$n \in \{1, 2\}.$$